# Base PLC-M12

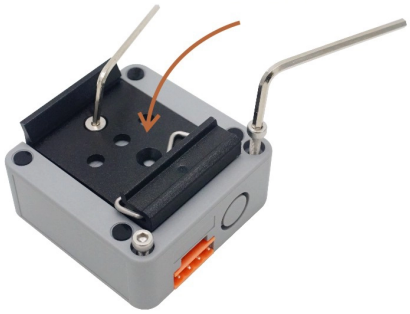
SKU:K011-B







For Din-Rail



For Magnet




## Description

**Base PLC-M12** is an industrial control base. It comes with an industrial-grade enclosure, designed specifically for industrial control applications. Unlike typical industrial PLCs (Programmable Logic Controllers), it has a more compact design. While delivering core industrial control functions, it also maintains low power consumption. Based on the M5 development ecosystem, users can add any number of relay switches as needed, and the base also supports TTL/RS485 communication. It supports functional module stacking expansion, for example, stacking a LoRa module can make the PLC a LoRa communication node. Additionally, the base includes a PLC - Proto board, making it easy for follow-up circuit development. With different hardware stacking and expansion designs, it can meet various customized needs in industrial-grade application scenarios. Therefore, **Base PLC-M12** is an ideal solution for industrial control projects.

## Features

- Voltage input range: 9 ~ 24V

## Includes

- 1 x PLC Proto Board
- 1 x PLC Base Enclosure
- 1 x RS485-to-TTL Converter Board
- 1 x HT3.96-7P Terminal

- 1 x HT3.96-4P Terminal
- 1 x Hex Key L-Shape 1.5mm
- 1 x Hex Key L-Shape 2.0mm
- 1 x Hex Key L-Shape 2.5mm
- 10 x 2mm Crimp Terminal
- 1 x Magnet (Center Hole, 15mm Diameter, 3mm Thickness)
- 1 x 35mm Silver Metal DIN Rail
- 1 x 35mm Black DIN Rail Clip
- 1 x M3 * 6mm countersunk head machine screw
- 2 x M3 * 28mm cap head machine screw
- 4 x M2 * 5mm cap head self-tapping screw
- 1 x Cable Gland (M12)
- 1 x 2.54mm-20P Straight Male Header (Overall Height 5.32mm)
- 1 x Product Sticker

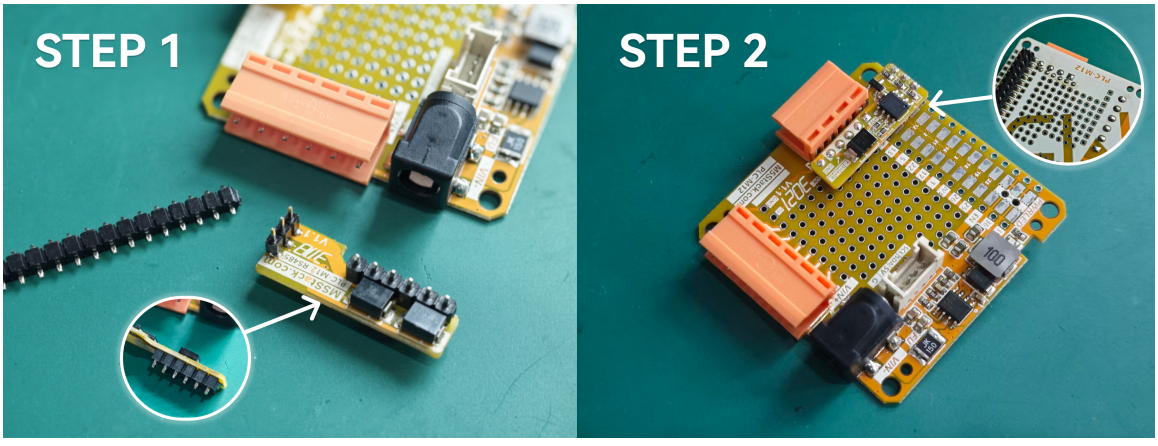
## Specifications

| Specification | Parameter |
|---|---|
| Product Weight | 25g |
| Gross Weight | 133g |
| Product Size | 54 x 54 x 28mm |
| Package Size | 105 x 65 x 40mm |

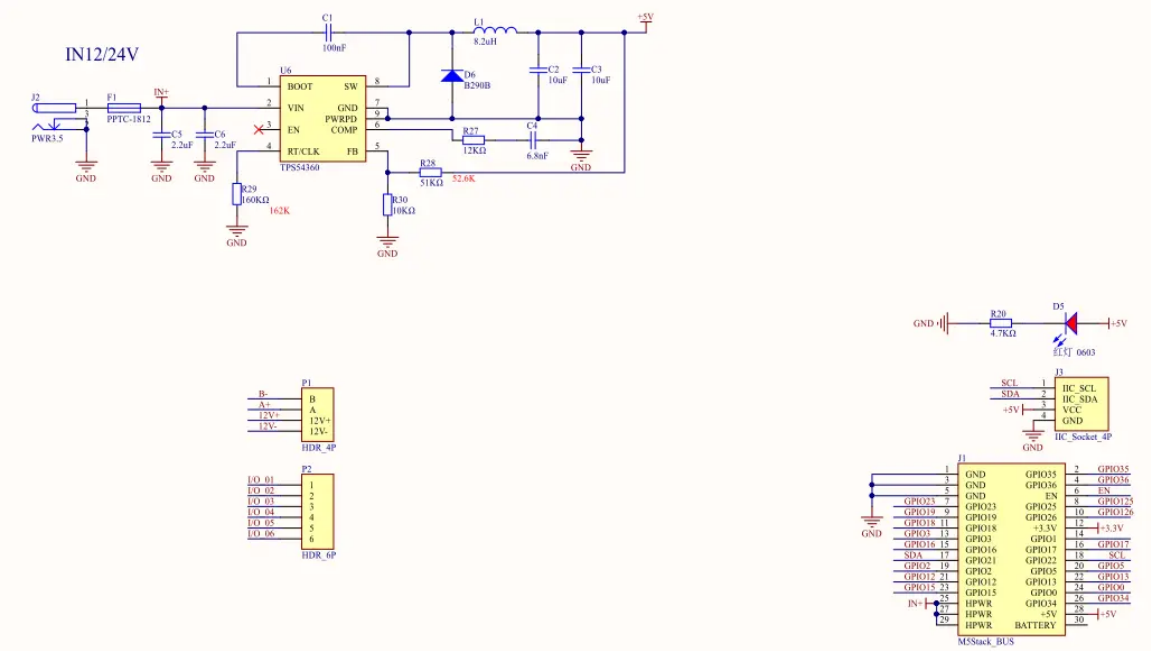
## Learn

If you need to add an RS485 communication interface, please solder the TTL-RS485 adapter board and its matching pin header to the corresponding pins on the motherboard. The serial port pins of the TTL-RS485 adapter board will connect to G16 and G17 of the PLC base.

### RS485 Expansion Board Installation




## Schematics



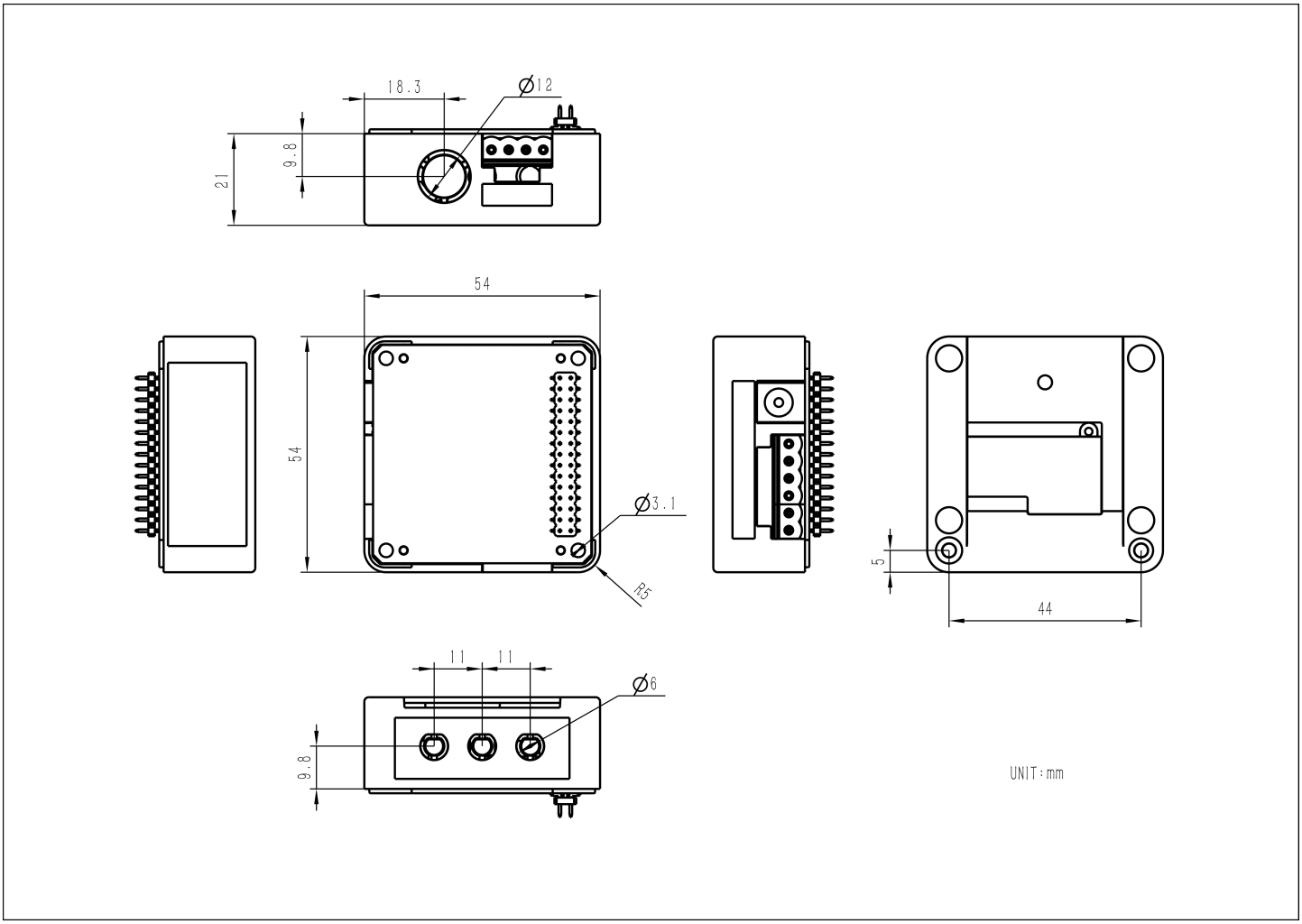

# PinMap

## M5-Bus

| PIN | LEFT | RIGHT | PIN |
|---|---|---|---|
| GND | 1 | 2 | NC |
| GND | 3 | 4 | NC |
| GND | 5 | 6 | NC |
| NC | 7 | 8 | NC |
| NC | 9 | 10 | NC |
| NC | 11 | 12 | 3V3 |
| NC | 13 | 14 | NC |
| RS485_TX | 15 | 16 | RS485_RX |
| NC | 17 | 18 | NC |
| NC | 19 | 20 | NC |
| NC | 21 | 22 | NC |
| NC | 23 | 24 | NC |
| HPWR | 25 | 26 | NC |
| HPWR | 27 | 28 | 5V |
| HPWR | 29 | 30 | BAT |

# Model Size

- Base PLC-M12 Model Size PDF




## Video

**RS485 Communication**

RS485 Application.mp4

## Version Change

| Release Date | Change Description | Note |
|---|---|---|
| / | First Release | / |
| 2023 | Upgraded to v1.1 | Updated accessories |